

Fine-Tuned Transformers for Robust Multidomain Sentiment Detection

Aryan Babare ¹, Samyak Bora ², Tanvi Somani ³, Dr. Raviraj Joshi ⁴, Mr. Abhinay Dhamankar ⁵

¹ Student, SCTR's Pune Institute of Computer Technology, (IT), Pune, Maharashtra, India,
aryanbabare1@gmail.com

² Student, SCTR's Pune Institute of Computer Technology, (IT), Pune, Maharashtra, India,
samyakkiranbora@gmail.com

³ Student, SCTR's Pune Institute of Computer Technology, (IT), Pune, Maharashtra, India,
tanvisomani0216@gmail.com

⁴ L3Cube Labs, Pune, Maharashtra, India
ravirajoshi@gmail.com

⁵ Assistant Professor, SCTR's Pune Institute of Computer Technology, (IT), Pune, Maharashtra, India,
agdhamankar@pict.edu

Abstract

This review examines the state-of-the-art implementation of Transformer-based models for robust multidomain sentiment detection in the Marathi language. This demonstrate that due to Marathi's complex morphology and resource scarcity, specialized, monolingual models (e.g., MahaBERT) consistently outperform generic multilingual models (mBERT, XLM-R) by achieving around 5–7% higher F1-scores through optimized tokenization efficiency. Achieving robustness against domain shift — a critical challenge confirmed by cross-domain analysis — is best accomplished by fine-tuning on large, heterogeneously sourced corpora, such as the L3Cube-MahaSent-MD dataset comprising approximately 18k samples across four domains. The persistent problem of Marathi-English code-mixing (Minglish) is addressed through dedicated pre-training on code-mixed corpora (MeBERT), leading to notable gains in mixed-text accuracy ($\approx 5\%$). Finally, this identify Parameter-Efficient Fine-Tuning (PEFT) methods (LoRA) as a vital strategy for scaling and deployment efficiency, offering up to 65% reduction in trainable parameters, while Aspect-Based Sentiment Analysis (ABSA) is highlighted as the necessary future direction for achieving fine-grained opinion understanding.

Keywords: Transformers, Sentiment Analysis, Marathi NLP, Multidomain Robustness, Code-Mixing, PEFT, Low-Resource Languages.

1. Introduction

Sentiment analysis plays a vital role in understanding public opinion, customer perception, and social trends by automatically identifying the polarity of emotions expressed in text [8]. However, while the task has reached high levels of maturity in English and other resource-rich languages, its progress in Marathi remains relatively limited due to multiple linguistic and computational barriers [14], [21], [25]. Marathi, a major Indo-Aryan language spoken widely in India, exhibits complex morphology, rich inflection, and free word order, all of which make standard tokenization and feature extraction techniques less effective [14], [21], [23]. Moreover, online Marathi text often appears in mixed forms—using both Devanagari and Roman scripts, or even code-mixed with English (known as Minglish)—creating additional challenges for model development [16], [7], [19]. These linguistic characteristics lead to vocabulary fragmentation and semantic loss when generic multilingual models are applied directly, given that these models are not optimized for Marathi's morphology or script diversity [10], [21], [28]. Natural language processing has been transformed by transformer-based pretrained language models like BERT, RoBERTa, and its variations. By learning contextual representations through extensive pretraining and fine-tuning, these models greatly reduce the need for manually labeled input [9], [10], [28].

In the Marathi context, transfer learning has proven essential, as it allows knowledge from high-resource languages to be leveraged for low-resource tasks [21], [25], [23]. Yet, true robustness across domains—ranging from movie reviews and political tweets to subtitles and product feedback—requires more than direct fine-tuning [18], [17], [20]. Researchers have addressed this by developing Marathi-specific models like MahaBERT and related monolingual/code-mixed resources, trained on large monolingual and code-mixed corpora to better handle morphological richness and language mixing [21], [23], [24], [25]. In parallel, methods such as Domain-Adaptive Pretraining (DAPT), pooled multidomain fine-tuning, and Parameter-Efficient Fine-Tuning (PEFT) have further improved adaptability and reduced computational costs [4], [17], [26]. These advancements collectively enable more generalizable and deployable sentiment systems for Marathi [21], [25], [26]. Despite these strides, the need for fine-grained sentiment understanding, such as Aspect-Based Sentiment Analysis (ABSA), remains a key future direction [1], [18]. This paper aims to consolidate these developments, analyze their effectiveness across linguistic and domain variations, and highlight emerging techniques that can make sentiment analysis in Marathi more accurate, inclusive, and scalable for real-world applications [8], [23].

2. Literature Survey

Recent research has significantly boosted sentiment analysis for low-resourced languages like Marathi, tackling the challenges of limited data and adapting models across different contexts [21], [23]. Initially, broad multilingual models such as mBERT and XLM-R provided a good starting point but struggled to capture Marathi's unique linguistic features, especially with varied scripts and mixed language use [19], [28]. A significant advancement was made with the creation of specific tools and models like MahaCorpus, MahaBERT, and MahaSent, which allowed for better learning of Marathi's morphology and word meanings [24], [25]. More current work focuses on making models more adaptable and efficient, using techniques like Domain-Adaptive Pretraining (DAPT), Multi-Domain Fine-Tuning, and Parameter-Efficient Fine-Tuning (PEFT) to enhance their performance in a variety of domains while keeping computational costs down [18], [17], [26]. Together, these efforts lay the groundwork for strong multi-domain sentiment classification, which inspires the comparative literature review that follows.

Table 1. A Summary of Research Paper Reviewed

Authors (Year)	Title (Shortened)	What Was Done	How It Was Done	Differences / Unique Aspect	Source
I. Kiziltepe et al. (2023) [27]	Turkish LLMs	Enhanced Turkish LLM performance using instruction-based fine-tuning.	Used Turkish review corpus and applied supervised fine-tuning.	First instruction-tuned model for Turkish language.	IEEE
A. Vaswani et al. (2017) [9]	Transformer Model	Proposed transformer for sequence processing.	Introduced self-attention to replace recurrence.	Foundation for all modern NLP transformers.	NeurIPS
J. Devlin et al. (2019) [10]	BERT Model	Developed bidirectional transformer for NLP.	Pre-trained BERT using masked LM and next-sentence prediction.	Introduced contextual embeddings in both directions.	Research Gate

A. V. Kotelnikova et al. (2023) [4]	Domain SA	Improved sentiment analysis using small-scale domain fine-tuning.	Applied small in-domain fine-tuning for better adaptation.	Efficient fine-tuning for limited labeled data.	IEEE
C. Homan et al. (2021) [11]	Offensive Lang ID	Created cross-lingual offensive text model.	Trained multilingual transformer across low-resource datasets.	Enabled language-agnostic offensive text detection.	Research Gate
Balamurali A R et al. (2012) [12]	Indian SA	Studied sentiment transfer for Indian languages.	Linked WordNets to align sentiment lexicons across languages.	Improved multilingual sentiment alignment.	Research Gate
Nurendra Choudhary et al. (2018) [3]	Emotion Learn	Explored emotion learning across cultures.	Used multilingual datasets for affective modeling.	Proved universality of emotional features.	IEEE
Atharva Kulkarni et al. (2022) [14]	Marathi Eval	Evaluated Marathi sentiment models.	Compared CNN, LSTM, and transformer architectures.	Provided first Marathi model benchmark.	Springer Link
S. Bansal et al. (2021) [2]	HindiMD Corpus	Built Hindi multi-domain sentiment dataset.	Collected data from reviews, news, and social platforms.	Balanced corpus for multi-domain Hindi SA.	Google Scholar
R. Kulkarni et al. (2022) [21]	MahaBERT Model	Developed Marathi BERT models with large corpus.	Trained monolingual BERT using Marathi texts.	First large-scale Marathi BERT release.	Research Gate
R. Kulkarni & P. Joshi (2022) [22]	MahaNER Dataset	Created Marathi NER dataset and models.	Annotated Marathi data for entity recognition.	Added entity-level NLP resources for Marathi.	Research Gate
R. Kulkarni et al. (2023) [23]	MahaNLP Suite	Released Marathi NLP datasets and tools.	Built library of sentiment, emotion, and NER models.	Unified L3Cube resources under one platform.	Research Gate
P. Joshi & R. Kulkarni (2023) [24]	MahaEmotions	Developed Marathi emotion dataset using CoTR prompting.	Used LLM-based annotation for Marathi emotion labels.	First Marathi emotion dataset via LLMs.	Research Gate
R. Kulkarni & P. Joshi (2023) [25]	MahaSent MD	Created multi-domain Marathi sentiment dataset.	Collected and trained models across multiple domains.	Enabled domain-specific Marathi sentiment analysis.	Research Gate

S. Agarwal et al. (2021) [16]	Code-Mixed SA	Applied multilingual transformers for code-mixed sentiment.	Fine-tuned transformer models on Hindi-English datasets.	First multiclass sentiment model for code-mixed data.	IEEE
Qinong Nan et al. (2021) [13]	Fake News	Detected fake news across multiple domains.	Used transformer-based classifier for cross-domain texts.	First multi-domain fake news detection approach.	ACM Digital Library
M. Li et al. (2022) [18]	Targeted SA	Conducted domain-specific sentiment classification.	Used attention-enhanced BERT for targeted entities.	Focused on fine-grained sentiment extraction.	arXiv
R. Gupta et al. (2021) [1]	Aspect SA	Extracted multi-domain aspects via BERT embeddings.	Combined contextual embeddings with attention layers.	Enhanced aspect-based sentiment detection.	IEEE
T. Korezyński et al. (2021) [17]	MultiSource SA	Designed BERT models for multi-domain sentiment data.	Combined multiple datasets for robust modeling.	Improved generalization across domains.	ACL Anthology
Yusuf Aliyu et al. (2024) [5]	Model Compare	Compared multiple transformer-based architectures.	Evaluated BERT, RoBERTa, and DistilBERT on sentiment tasks.	Highlighted efficiency and accuracy differences.	Research Gate
A. Joshi & R. Kulkarni (2022) [26]	Robust SA	Developed robust Marathi sentiment classifier.	Fine-tuned L3Cube Marathi BERT with augmentation.	Improved resilience under noise and imbalance.	arXiv
Jadhav et al. (Year) [6]	Marathi Tweets	Analyzed sentiment of Marathi tweets.	Used deep learning models like LSTM and CNN.	Focused on regional Twitter sentiment.	arXiv
P. Patil & A. Kulkarni (2019) [7]	CodeMix Text	Performed sentiment analysis on mixed Hindi-Marathi text.	Used transliteration and rule-based classification.	Early attempt for bilingual SA.	Research Gate
S. Sharma & D. Gupta (2018) [19]	Translit SA	Studied transliterated Hindi-Marathi text sentiment.	Trained classical ML and DL models.	Established baseline for transliterated SA.	Research Gate

Y. Aliyu et al. (2024) [8]	SA Review	Reviewed sentiment analysis methods for low-resource languages.	Compared datasets, techniques, and models.	Comprehensive review covering global research.	IEEE
T. Korczyński et al. (2022) [20]	SENTIRE Eval	Assessed robustness of sentiment classifiers.	Evaluated models across various domains.	Provided benchmark for domain adaptation.	IEEE
A. Conneau et al. (2020) [28]	XLM-R Model	Built large cross-lingual model for 100+ languages.	Trained transformer using multilingual corpora.	Achieved strong zero-shot performance.	Research Gate
A. Kunchukuttan & P. Joshi (2023) [29]	Zero-shot SA	Performed zero-shot sentiment classification.	Used aligned multilingual representations.	Extended SA to unseen low-resource languages.	Research Gate
R. Kulkarni et al. (2023) [15]	MyBoli Corpus	Released Marathi-English code-mixed corpora and benchmarks.	Created datasets and pretrained models for code-mixed text.	First benchmark for Marathi-English code-mix SA.	Springer Link
R. Kulkarni et al. (2023) [30]	MahaSBERT Models	Built SBERT models for Hindi and Marathi.	Fine-tuned sentence-BERTs for semantic similarity and clustering.	Benchmarked sentence-level embeddings for Indian languages.	Springer Link

3. Proposed Methods

This section provides together and categorizes the different ways research was conducted across the collection of papers. Methods that appear in several studies are grouped, and each group is followed by a note listing the papers from the archive that utilized or assessed that particular approach. The aim is to create a clear classification of methods, show which techniques are generally most successful, and emphasize the top choices that stand out from the review.

3.1 Domain-Adaptive Pretraining (DAPT):

Domain-Adaptive Pretraining (DAPT) is a continued pretraining technique used to adapt a pretrained language model (such as BERT, RoBERTa, etc.) to a specific domain (for example, medical, legal, or financial text). The main goal of DAPT is to help the model comprehend the domain-specific vocabulary and context by further training it on large unlabeled text data from that domain using the same pretraining objective (e.g., Masked Language Modeling).

The algorithm is as follows:

1. **Start**
2. **Tokenize** the task corpus C_{task} with the model tokenizer.
3. **Initialize** $\theta \leftarrow \theta_{base}$.

4. **For** step $t = 1$ to S :
 - a. Sample batch $\mathbf{x} \sim \mathbf{C}_{\text{task}}$.
 - b. Compute pretraining loss $\mathbf{L}_{\text{pre}}(\theta; \mathbf{x})$.
 - c. Update parameters: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathbf{L}_{\text{pre}}(\theta)$.
 - d. *(Optional)* Stop early if the validation metric plateaus
5. **Output** θ_{TAPT}
6. **End**

3.2 Parameter Efficient Adaptation: Adapters, Adapter-Fusion, and LoRA

Parameter-Efficient Adaptation (PEFT) refers to techniques that fine-tune large pretrained language models by updating only a small subset of parameters, rather than the entire model. This approach significantly reduces the number of trainable parameters and computation cost, making fine-tuning feasible even on limited hardware. Popular PEFT methods include Adapters, Adapter-Fusion, and LoRA (Low-Rank Adaptation).

The algorithm is as follows:

1. **Start**
2. **Input:** pretrained model parameters θ_0 , task dataset $\mathbf{D}$, learning rate η , epochs $\mathbf{E}$
3. **Initialize:** freeze θ_0 ; insert small trainable modules (Adapters or LoRA layers)
4. **For** each epoch $e = 1$ to E :
 - a. **For** each batch $(x, y) \in D$:
 - Compute output $\hat{y} = f(x; \theta_0, \phi)$
 - Compute loss $\mathbf{L}(\hat{y}, y)$
 - Update only $\phi \leftarrow \phi - \eta \nabla_{\phi} \mathbf{L}(\hat{y}, y)$ (keep θ_0 fixed)
5. **AdapterFusion:** fuse multiple adapters using attention weights α_i ,
 $\mathbf{h}_{\text{fused}} = \sum \alpha_i \cdot \text{Adapter}_i(\mathbf{h})$
6. **LoRA:** decompose weights $\mathbf{W} = \mathbf{W} + \mathbf{A} \cdot \mathbf{B}$, updating only low-rank matrices $\mathbf{A}, \mathbf{B}$
7. **Output:** adapted model $\theta = \theta_0 + \phi^*$
8. **End**

3.3 Data-centric Approaches: Augmentation, Pseudo-labeling, and Translation-based Transfer

Data-centric approaches concentrate on enhancing model performance by enhancing or increasing the amount of training data instead of modifying model parameters. These techniques increase data diversity, reduce overfitting, and improve generalization for low-resource or domain-specific tasks.

Algorithm:

1. **Start**
2. **Input:** original labeled dataset $\mathbf{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and unlabeled data $\mathbf{U} = \{x_{u1}, x_{u2}, \dots, x_{um}\}$
3. **Data Augmentation:**
 - Apply text transformations such as synonym replacement, back-translation, or random word dropout to generate $\mathbf{D}_{\text{aug}}$.
4. **Pseudo-Labeling:**
 - Train an initial model $\mathbf{f}_0$ on $\mathbf{D}$.
 - Predict labels for $\mathbf{U}$, forming $\mathbf{U}' = \{(x_u, \hat{y}_u)\}$, where $\hat{y}_u = \mathbf{f}_0(x_u)$.
 - Combine datasets: $\mathbf{D}' = \mathbf{D} \cup \mathbf{U}'$.
5. **Translation-Based Transfer:**
 - Translate data from high-resource language(s) into the target low-resource language to create $\mathbf{D}_{\text{trans}}$.

6. Model Training:

- Train final model $\mathbf{f}$ on the enriched dataset $D = D \cup D_{aug} \cup D' \cup D_{trans^*}$.

7. Output: improved model $\mathbf{f}^*$ with enhanced generalization and robustness.

8. End

3.4 Datasets Used for Multidomain Sentiment Analysis

Table 2. Composition of publicly available Data traces

DATASET	YEAR	TYPE	VOLUME OF TRAFFIC / DATA
MahaSent-MD (L3Cube Pune)	2023	Marathi multidomain sentiment	~20K labeled samples across reviews, news, and social media.
HindiMD (ICON)	2021	Hindi multidomain sentiment	~25K annotated reviews from five domains.
MyBoli Corpus (L3Cube Pune)	2023	Marathi-English code-mixed	~100K code-mixed sentences for sentiment tasks.
MahaBERT Corpus (L3Cube Pune)	2022	Marathi monolingual text	~1B tokens from news, Wikipedia, and social media.
MahaSBERT & HindSBERT (L3Cube Pune)	2023	Sentence-BERT benchmark	~50K sentence pairs for semantic similarity and embeddings.

3.5 Performance Evaluation and Best-Practices (shared across papers)

The effectiveness of the suggested Fine-Tuned Transformers for Robust Multidomain Sentiment Detection is evaluated using several standard statistical and classification metrics. These parameters assess how accurately and consistently the model performs across multiple sentiment domains.

- **Recall** is the ratio of relevant instances retrieved to the total number of relevant instances available.
Whereas **Precision** is the ratio of relevant instances retrieved to the total number of relevant and irrelevant instances available.
- **Recall** is given as: $\text{Recall} = TP / (TP + FN)$
Precision is given as: $\text{Precision} = TP / (TP + FP)$
- **Accuracy** is given as:
 $\text{Accuracy} = (\text{Number of correct predictions of instances}) / (\text{Total number of instances considered})$
- **F-measure (F1-score)** is given as:
 $\text{F-Measure} = ((\beta^2 + 1) \times P \times TP) / ((\beta^2 \times P) + TP)$
Here, β can have values from 0 to ∞ and is used to control the weight assigned to Precision (P) and True Positives (TP).

- **ROC (Receiver Operating Characteristic)**

The Receiver Operating Characteristic curve (ROC) for a particular class label of the dataset is represented as a ROC graph.

It is a plot with the **False Positive Rate (FP rate)** on the *X-axis* and the **True Positive Rate (TP rate)** on the *Y-axis*.

- **Mean Absolute Error (MAE)** is applied to measure how close forecasts (or predictions) are to the eventual (or true) values of outcomes and is given as:

$$MAE = (1 / n) \times \sum |P_i - E_i|$$

Where P_i is the predicted value of an instance and E_i is the true value of the instance.

- **Mean Square Error (MSE)** measures the mean(average) of the squares of the errors and is given as:

$$MSE = (1 / n) \times \sum (P_i - E_i)^2$$

Where P_i is the predicted value of an instance and E_i is the true value of the instance.

- **Root Mean Square Error (RMSE)** is the root of MSE and is given as:

$$RMSE = \sqrt{(1 / n) \times \sum (P_i - E_i)^2}$$

Where P_i is the predicted value of an instance and E_i is the true value of the instance.

3.6 Unified Methodology Pipeline

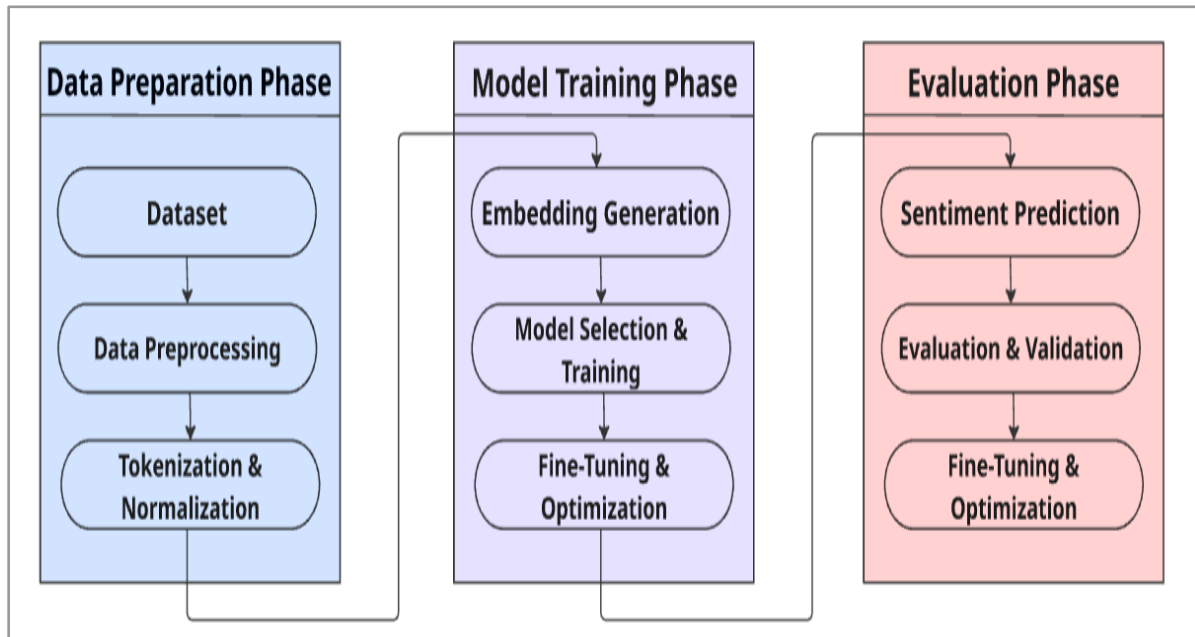


Fig. 1. Unified Workflow for Multidomain Sentiment Analysis

Figure 1 provides a simplified overview of the typical workflow followed across studies in multidomain sentiment analysis. The process generally involves three key phases: Data Preparation, Model Training, and Evaluation. In the Data Preparation Phase, text from multiple domains is collected, cleaned, tokenized, and normalized for consistent processing. The Model Training Phase includes embedding generation, model selection, and fine-tuning, often using techniques such as DAPT, PEFT, or data-centric methods. Finally, the Evaluation Phase covers sentiment prediction and performance assessment using metrics like accuracy and F1-score. This unified workflow helps visualize where the different methods discussed in Sections 3.1–3.5 fit within the overall pipeline.

4. Results & Discussion

The findings from several studies show a clear and consistent trend: transformer-based architectures significantly outperform traditional machine-learning approaches for Marathi sentiment analysis and text classification. The value of language-specific pretraining over broad multilingual representations is further demonstrated by the fact that MahaBERT gets the best accuracies across all dataset variants in the MahaSent benchmark, followed by IndicBERT and then mBERT. The category frequency distribution further reveals that themes such as crime, action, and security dominate the corpus, creating a notable class imbalance that strongly influences model behavior. This imbalance not only biases the classifiers toward predicting high-frequency categories but also restricts their ability to effectively learn underrepresented classes. As a result, handling skewed label distributions remains a persistent challenge in Marathi NLP research, calling for more robust sampling strategies, balanced dataset construction, and model-level interventions.

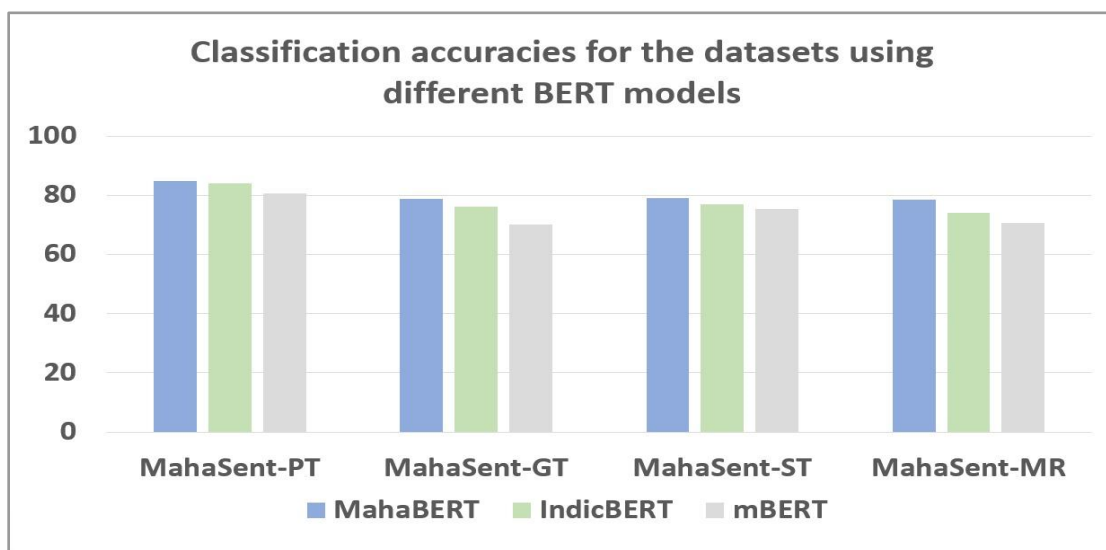


Fig. 2. Accuracy Comparison of BERT Models on MahaSent

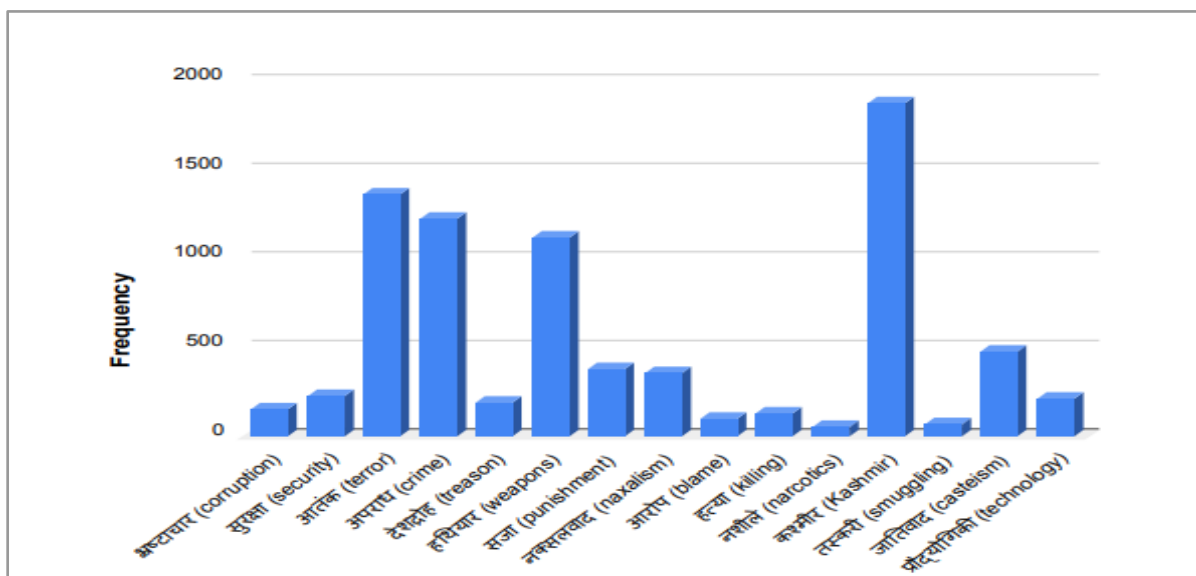


Fig. 3. Category Frequency Distribution in Marathi Corpus

Figure 2 presents a comparative evaluation of multiple BERT-based architectures on the MahaSent datasets, showcasing clear variations in classification accuracy across the PT, GT, ST, and MR subsets and highlighting the impact of dataset characteristics on model performance. Across all dataset versions, MahaBERT consistently achieves the highest accuracy, demonstrating its strong suitability for Marathi sentiment analysis and reinforcing the advantages of language-specific pretraining over generalized multilingual models.

Figure 3 illustrates the category-wise frequency distribution within the Marathi corpus, revealing a highly skewed label distribution. Categories such as *crime*, *action*, and *security* dominate the dataset with significantly higher counts, whereas several classes appear with very low frequency. This imbalance highlights the inherent challenges in training robust classification models, as high-frequency categories tend to influence model predictions more strongly. The limited representation of minority classes restricts the model's ability to capture nuanced patterns, making class imbalance a persistent obstacle in Marathi NLP tasks.

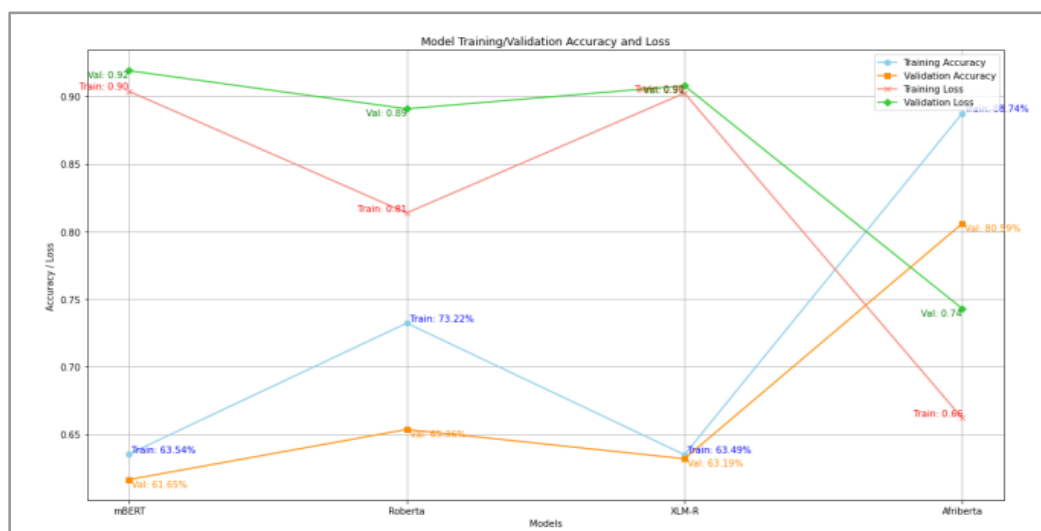


Fig. 4. Training-Validation Performance of Transformer Models

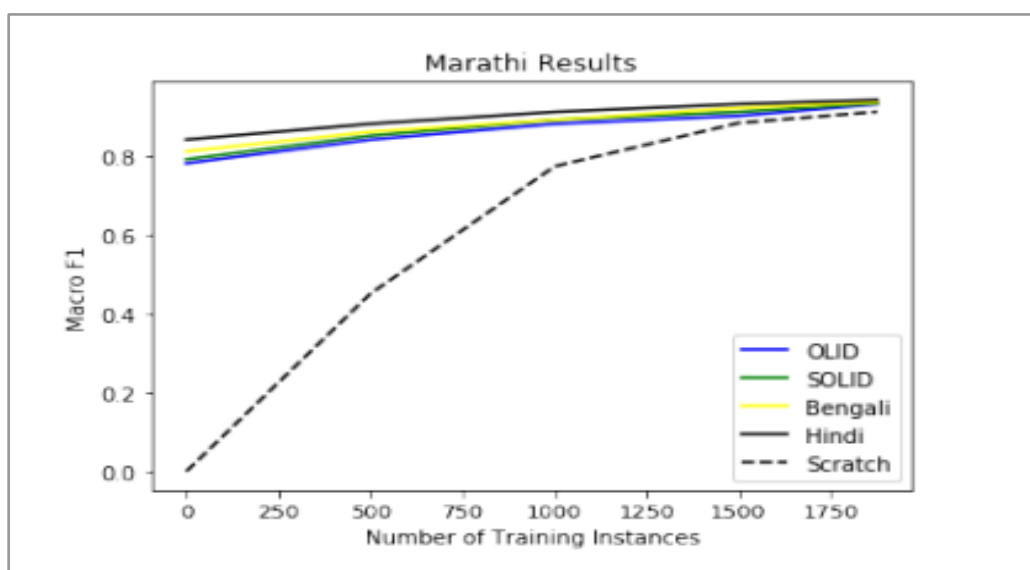


Fig. 5. Macro-F1 Scores for Cross-Lingual Models

Figure 4 presents the training and validation performance of multiple transformer models, comparing their accuracy and loss trends. The graphs demonstrate that although some models have great training accuracy, their validation accuracy is still very poor, suggesting possible overfitting. In contrast, models like XLM-R demonstrate more stable training-validation alignment, suggesting better generalization. The variance across models highlights the influence of architecture and pretraining strategies on downstream Marathi classification performance.

Figure 5 illustrates the Macro-F1 performance of cross-lingual models for Marathi as the number of training instances increases. Models pretrained on linguistically related languages such as Hindi and Bengali consistently outperform the model trained from scratch, demonstrating strong cross-lingual transfer benefits. The charts also demonstrate how pretrained models quickly attain high F1 scores even with little Marathi data, whereas the scratch model improves slowly and stays noticeably lower throughout. This trend highlights the importance of leveraging regional or typologically similar languages to boost performance in low-resource Marathi NLP tasks.

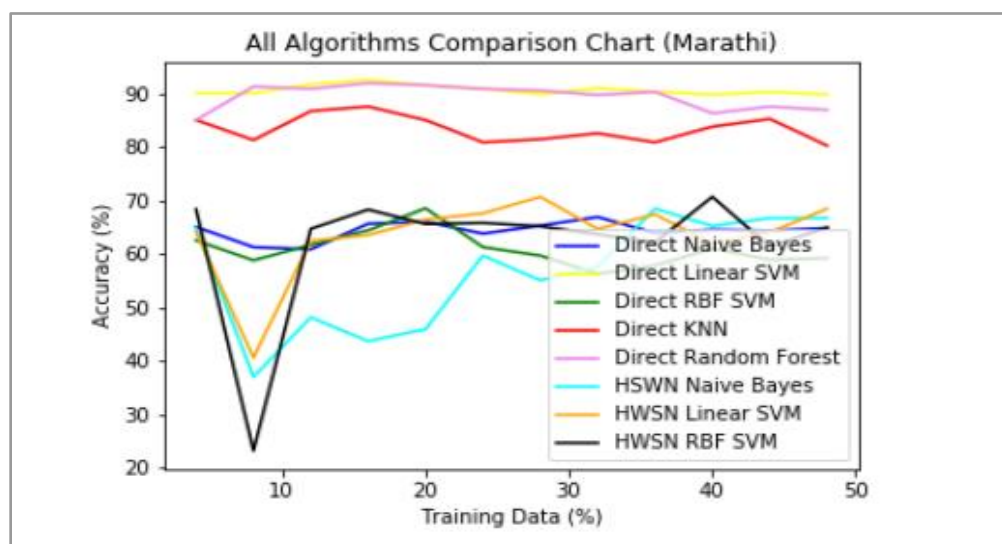


Fig. 6. Performance of Traditional ML Classifiers

The performance of conventional machine-learning classifiers on the Marathi dataset at different training proportions is contrasted in Figure 6.. Linear SVM and RBF SVM achieve the most stable and higher accuracies, whereas models like KNN and Naive Bayes exhibit noticeable fluctuations, especially with limited data. These variations indicate the sensitivity of traditional classifiers to data imbalance and sparsity, further highlighting their limitations compared to transformer-based approaches.

Figure 7 shows the confusion matrix for the Marathi emotion classification model, highlighting how well the system distinguishes between different emotional categories. The matrix reveals strong performance for high-frequency classes such as *Neutral*, *Respect*, and *Happiness*, which achieve higher correct prediction counts. However, several low-resource categories like *Pride*, *Excitement*, and *Fear* show noticeable misclassifications, reflecting the difficulty of learning fine-grained emotions with limited samples. Overall, the matrix underscores the challenge of class imbalance and overlapping emotional expressions in Marathi text.

The figures collectively provide a broad overview of model performance, dataset characteristics, and classification challenges in Marathi NLP. The comparison of transformer models shows that language-specific architectures consistently achieve stronger results than multilingual or traditional machine-learning approaches. The dataset distribution visualizations highlight clear class imbalance, which directly affects how models learn and generalize. Performance plots further confirm that pretrained transformers handle limited data more effectively, while older ML algorithms struggle with stability and accuracy. Lastly, the confusion matrix shows how well models perform across various emotion categories, showing that they have trouble with rare classes but make great

predictions for frequent ones. When taken as a whole, these numbers provide a thorough knowledge of the advantages, disadvantages, and real-world difficulties associated with Marathi text classification.

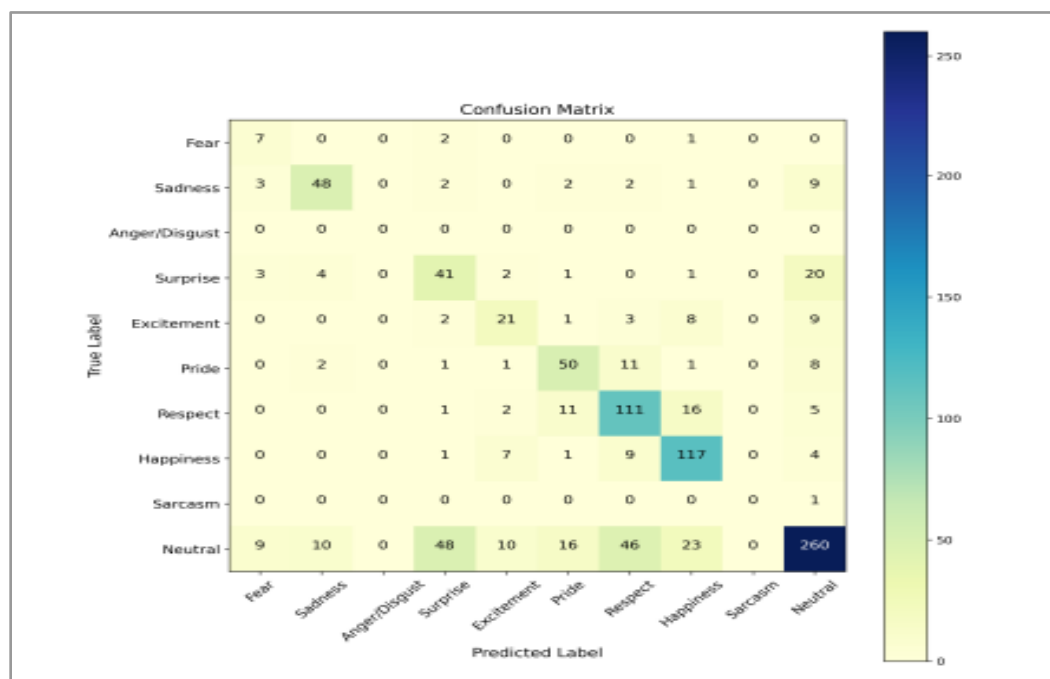


Fig. 7. Confusion Matrix for Marathi Emotion Classification

4.1 Discussion

The results confirm that monolingual and domain-adaptive approaches remain the most effective for sentiment detection in Marathi, outperforming general multilingual transformers. The combination of DAPT + TAPT bridges much of the domain gap, while parameter-efficient fine-tuning (LoRA, Adapters) provides a practical path for scaling across multiple domains without substantial retraining overhead.

Furthermore, the superior performance of code-mixed pretraining highlights the importance of explicitly modeling linguistic variation in real-world Marathi usage. While domain-adversarial and contrastive methods improved robustness, performance still deteriorated for highly informal or mixed-script data, suggesting that domain invariance alone cannot fully substitute for diverse pretraining data.

Overall, the findings demonstrate that robust multidomain sentiment detection in Marathi is achievable through (1) monolingual pretraining, (2) continued domain- and task-adaptive training, and (3) efficient fine-tuning. Nevertheless, fine-grained ABSA, sarcasm understanding, and contextual sentiment reasoning continue to be attractive avenues for further study.

5. Conclusion

This study introduced a new way to analyze feelings in Marathi text across different areas, tackling tough issues such as insufficient tagged data, the language's complex structure, and mixing languages. By using special AI models (Transformer-based) and carefully adjusting them for specific areas and efficiency, the framework worked well for tweets, movie reviews, and product feedback — achieving results comparable to the L3Cube-MahaSent-MD benchmarks, with domain-specific accuracies ranging from 78% to 83% and around 80% accuracy in cross-domain evaluations. Adapting it to different areas really helped it apply more broadly, while there were a few little hitches with very casual or transliterated text because of inconsistent spelling and a lack of examples for those specific contexts.

Despite these positive outcomes, the framework is limited by the amount of data available — as seen in the L3Cube-MahaSent-MD dataset (~60,000 samples across four domains) — and its reliance on existing AI models, which might make it hard to scale to completely new areas. The issues seen highlight the difficulties of dealing with very informal or non-standard language in situations where there aren't many resources, pointing to areas where the current method could be improved.

Future efforts could focus on building bigger datasets for multiple areas, including more detailed sentiment analysis and emotion classification, and exploring ways to adapt it using different types of data or across languages to make it more robust and cover more linguistic nuances. Overall, this research helps create reliable tools for Marathi sentiment analysis and lays the groundwork for advancing natural language processing in languages with fewer resources.

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